Computer Network I

Reti di Calcolatori I

Università di Napoli Federico II – Scuola Politecnica e delle Scienze di Base
Corso di Laurea in Informatica

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Introduction

Sharing Services and Resources

- The main goal of computer networks (and internet) is **to share services** or **resources** between different devices (and users beyond devices).

- There is a variety of **services** enabled by computer networks:
  - **File sharing**: moving, reading, copying files from remote hosts over the network.
  - **E-mails**: sending/receiving electronic mails.
  - **Instant messaging**: sending/receiving messages on-line.
  - **Video telephony**: organize and participate to calls.
  - **Web pages**: watching/showing hypertextual pages.
  - **Hardware management**: remote control (printing, surveillance, domotic systems, etc.).
  - **Hardware sharing**: using/providing hardware (CPU, memory, etc.) of/to remote hosts.
  - **Application sharing**: using/offering applications of/to remote hosts.
  - …

Introduction

Sharing Services on Local Networks

- Resources can be **private** and locally available: for instance, in local networks, there can be devices (printers, air conditioners, etc.) that offer services and devices (tablet, laptop, smartphone) that uses them.

- It could be possible to use such services (for instance, printing a document or turning on air conditioning) even **without internet access**.

- However, most of todays services are **public** and provided through Internet by Service Providers.



Introduction

Service Providers Beyond Connectivity

- There are several famous companies whose main business is to provide services over Internet:
  - Amazon (AWS),
  - Google,
  - Microsoft (Azure),
  - IBM,
  - Oracle,
  - Aruba,
  - Alibaba,
  - …

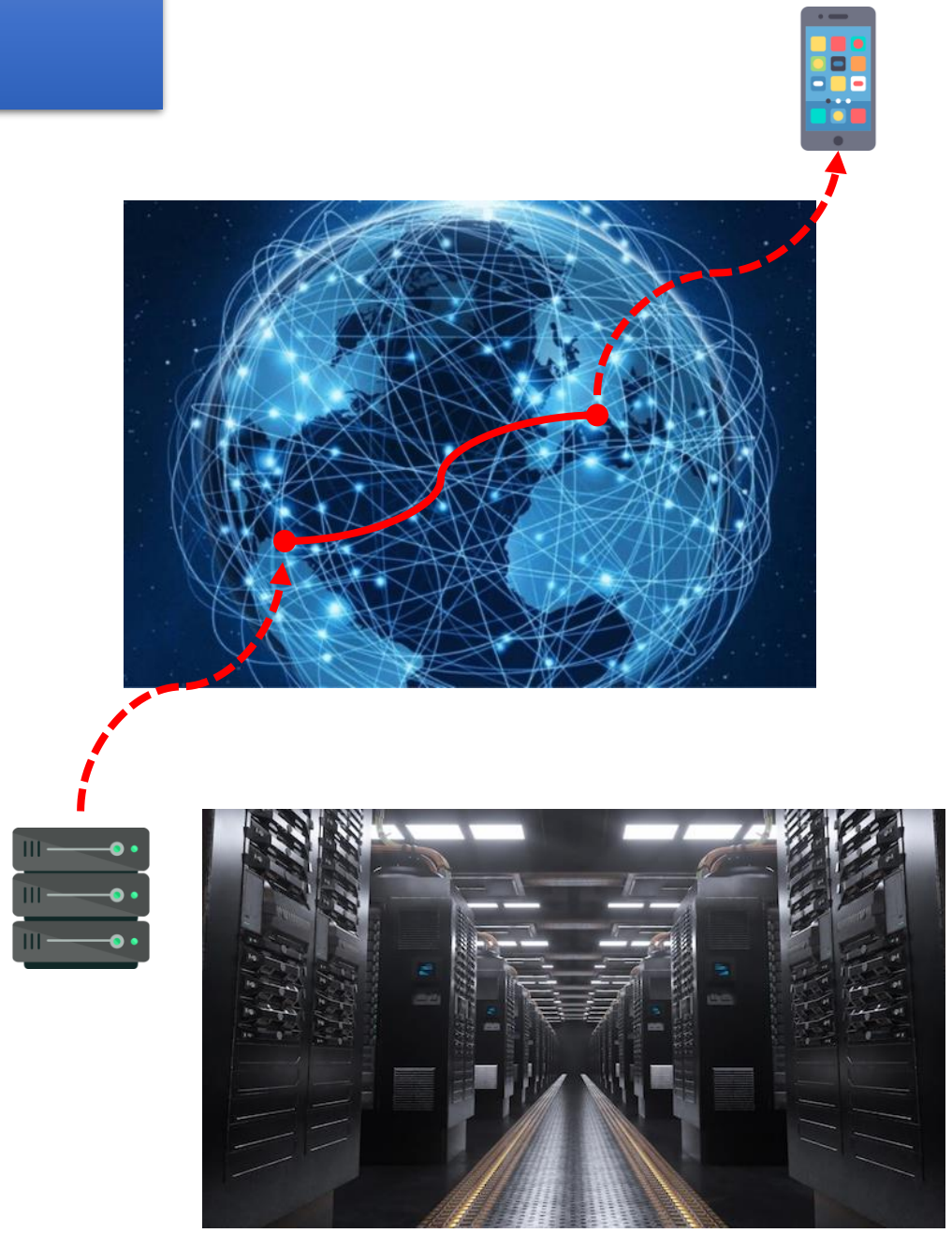


- There are different ways of providing services.

Introduction

Dedicated Servers

- Resources and services can be public and provided by distant **servers**.
  - A **server** is a special host designed to provide one or multiple **services**.
- For example, a web server (web pages) or an e-mail server may be on the other side of the world, but we are *almost unaware of this distance*.
- Moreover, there could be multiple servers cooperating to provide services and we would be again *unaware* of that.



Introduction
Grid Computing

- The **grid computing** is a decentralized resource-sharing infrastructure that typically combines hardware (resources) from different hosts in different geographic locations to achieve a common goal:
  - to perform complex calculations.
  - to store big quantities of data.

- A famous example is **SETI** (Search for Extra-Terrestrial Intelligence - 1999).

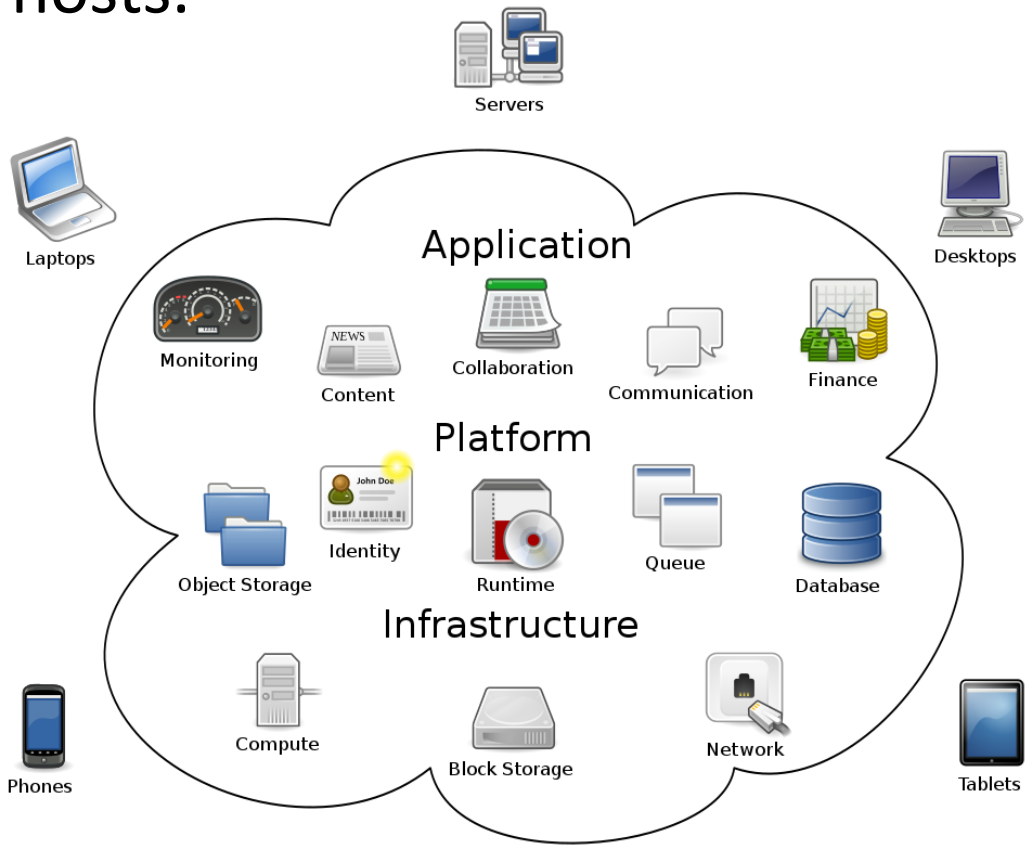
- The **SETI@home** project allows PCs all over the world shares resources to analyze radio signals from space, searching for extraterrestrial intelligence.



Introduction
Cloud Computing

- The **cloud computing** is centralized architecture in which resources (hardware and software), typically managed by companies (service providers), are offered on-demand as services to external hosts.

- In a cloud architecture several services can be offered:
  - Applications for productivity, games, communication, social networks, etc.
  - Storage, shared databases, web servers, etc.
  - Virtual machines (with specific configurations), servers, firewalls, etc.



Introduction

Cloud Computing

- A widely used definition of cloud computing is provided by the **NIST** (U.S. National Institute of Standards and Technology).

- **Cloud computing** is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

- This cloud model is composed of five essential characteristics:
  1. On-demand self-service.
  2. Broad network access.
  3. Resource pooling.
  4. Rapid elasticity.
  5. Measured service.

Introduction

Cloud Computing

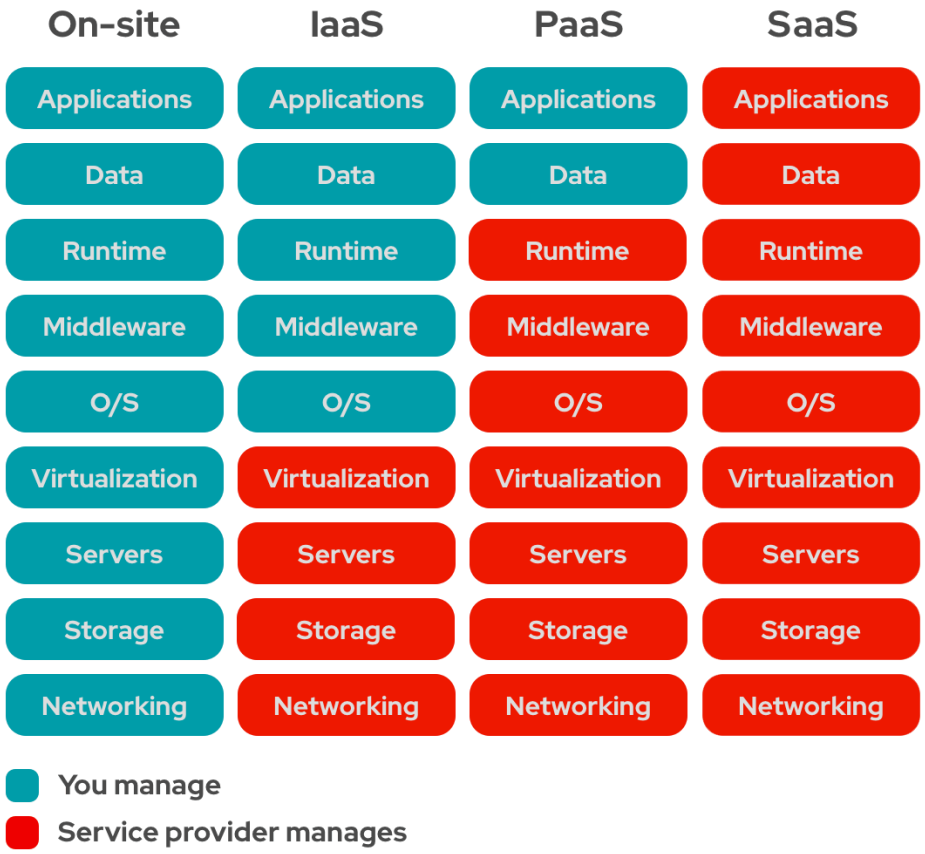
1. **On-demand self-service**. A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
2. **Broad network access**. Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
3. **Resource pooling**. The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
4. **Rapid elasticity**. Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.
5. **Measured service**. Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service. Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

Introduction

Cloud Computing: Service Models

- The NIST grouped services provided by a cloud computing architecture in 3 categories (**service models**) depending on how much of the computing infrastructure is managed by the service provider.

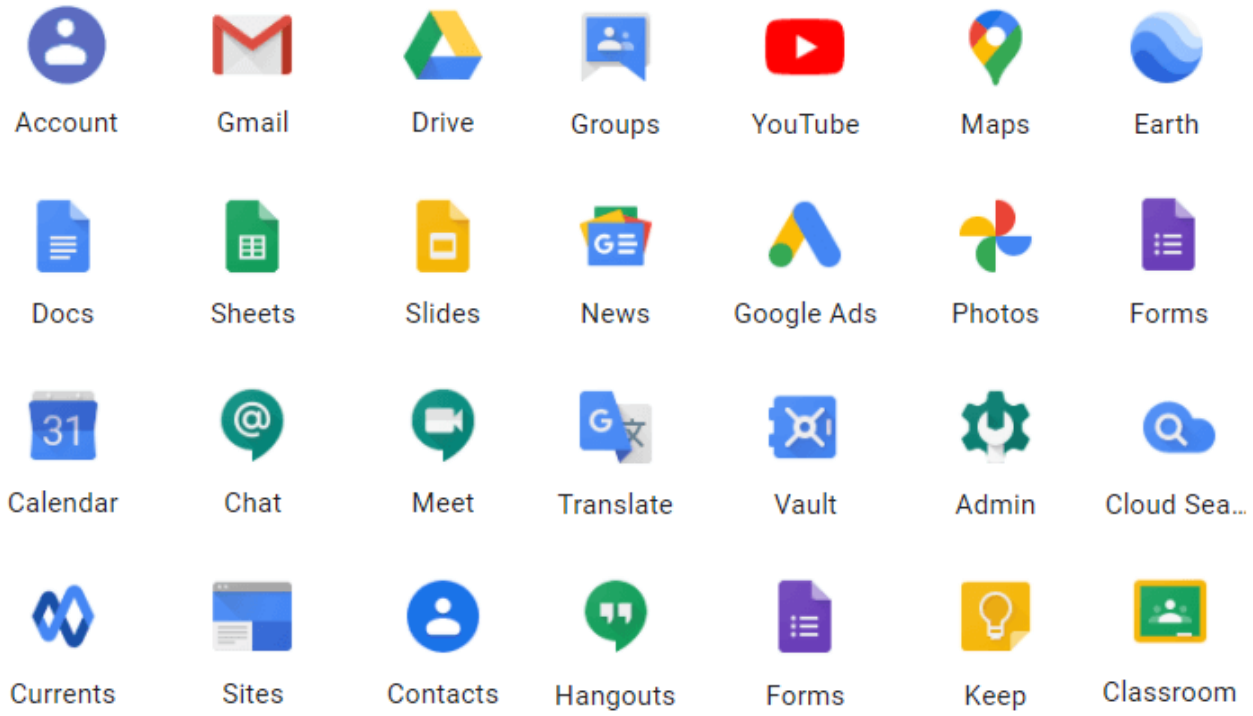
- **Software as a Service** (SaaS): whole applications provided along with software/hardware components needed to run them.
- **Platform as a Service** (PaaS): working platforms provided with specific configurations (e.g., OS, APIs, etc.).
- **Infrastructure as a Service** (IaaS): mainly the hardware is provided.



Introduction

Example of Cloud Service Provider

- Google is an example of Cloud Service Provider (CSP).
  - The most popular services are offered through the Google Workspace (mostly SaaS).
  - Additional services are available as part of the Google Cloud infrastructure (100+ available services).



Google Workspace



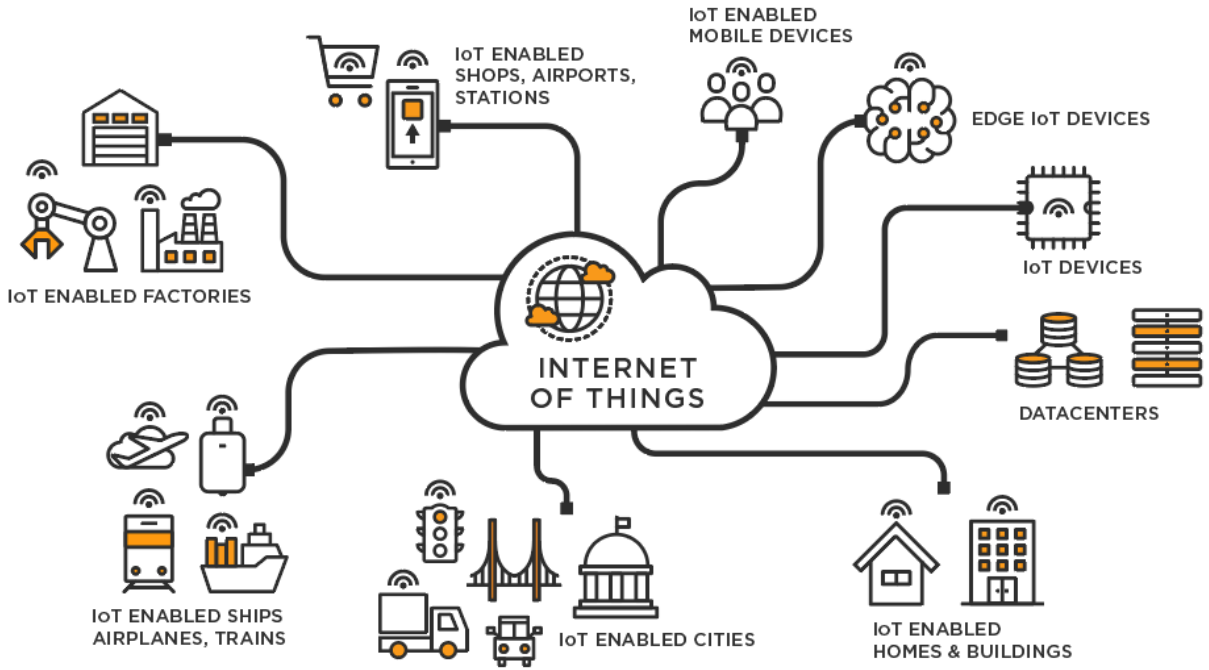
Groups of Google Cloud services

Introduction

Internet of Things

- The **internet of things** (IoT) is the approach of endowing simple devices (smart devices) with sensors, processing units, connectivity, etc. in order to control/monitor them remotely over internet.

- IoT can be used in combination with cloud computing to:
  - On-line manage devices using graphical user interfaces (GUIs) and remote applications.
  - Store data and information from devices.
  - Elaborate information by means of predictive algorithms or machine learning techniques.



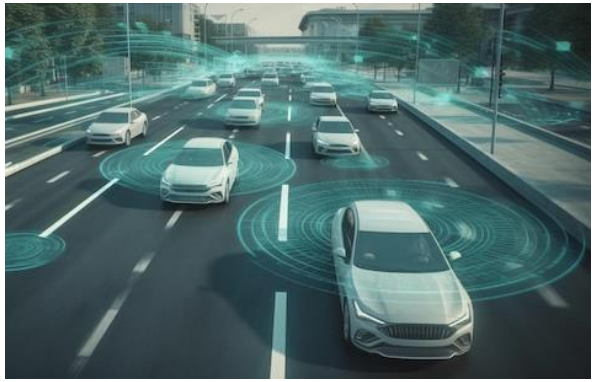
Introduction

Internet of Things

- Several “things” that are capable of internet connection have already been integrated in our lives:
  - Smartphones
  - Smartwatches
  - Home Assistants
  - Smart TV
  - Smart Cars
  - …











- Connected things particularly emphasize issues about *privacy* and *security*.

Introduction

Needs for Standards and Protocols

- Computer networks (such as Internet) can be quite **complex and widespread** over different locations, countries, users, etc.

- Clearly, a **set of rules** or a common ground should be defined so that all participants know how **to provide or to use services**.

- From the beginning of Internet, one of the major effort has been made to define **protocols** and **standards** that regulate communication.

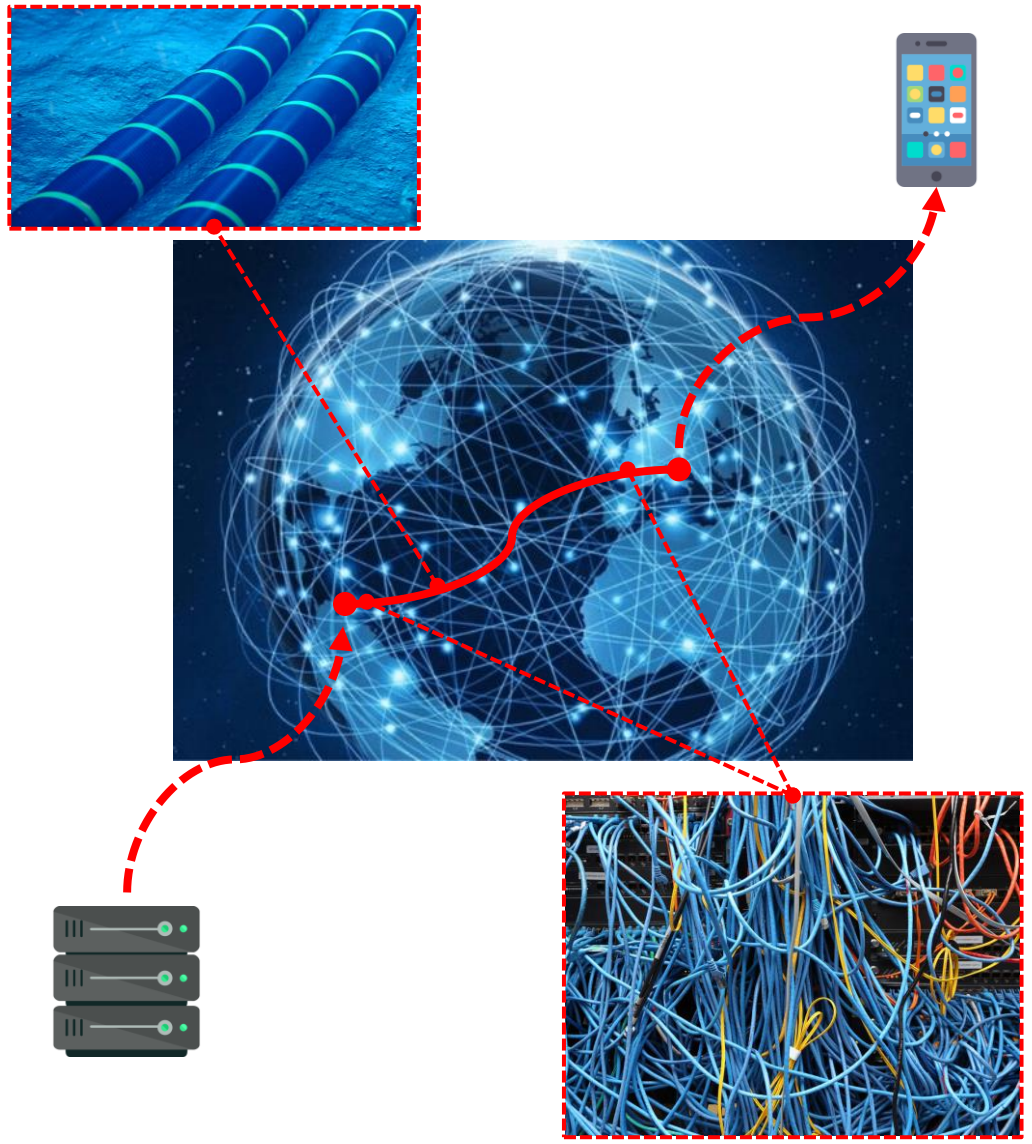


The Tower of Babel by Pieter Bruegel the Elder (1563).

Introduction

Needs for Standards and Protocols

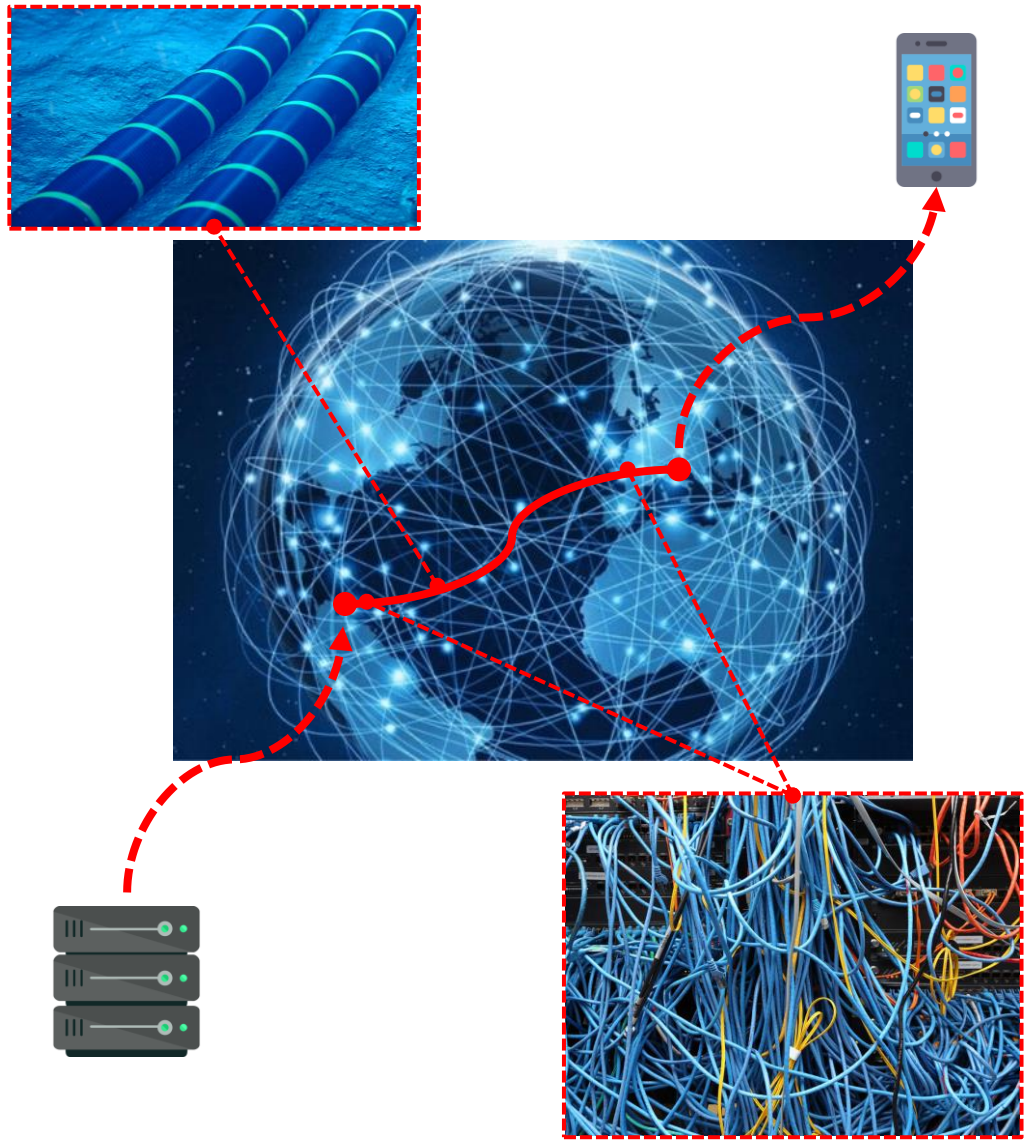
- The communication between devices involves **several problems**:
  - Physical transportation, addressing, error check, data conversion and regulation, security, synchronization, etc.
- To allow the communication both the receiver and the transmitter must agree to a **protocol**.
- When multiple/different devices from different places communicate they all must to agree on a common protocol, which becomes a **standard**.



Introduction

Needs for Standards and Protocols

- There are **hundreds of protocols** which regulate all aspects of communication from physical links to applications:
  - How **cables and links** should be created (materials, shielding, frequencies, etc.).
  - How **addresses** should be assigned.
  - How data should be wrapped into **packets or frames** for transmission.
  - How **errors** should be detected or corrected.
  - How **applications** should exchange data.
  - …



Introduction

Who Standardize Standards?

- Todays Internet standards and protocols are defined by a community of experts called **Internet Engineering Task Force** (IETF).



- IETF is typically organized in **open working groups**, each one focused on specific aspects of internet, whose members interacts by mailing list and meetings.

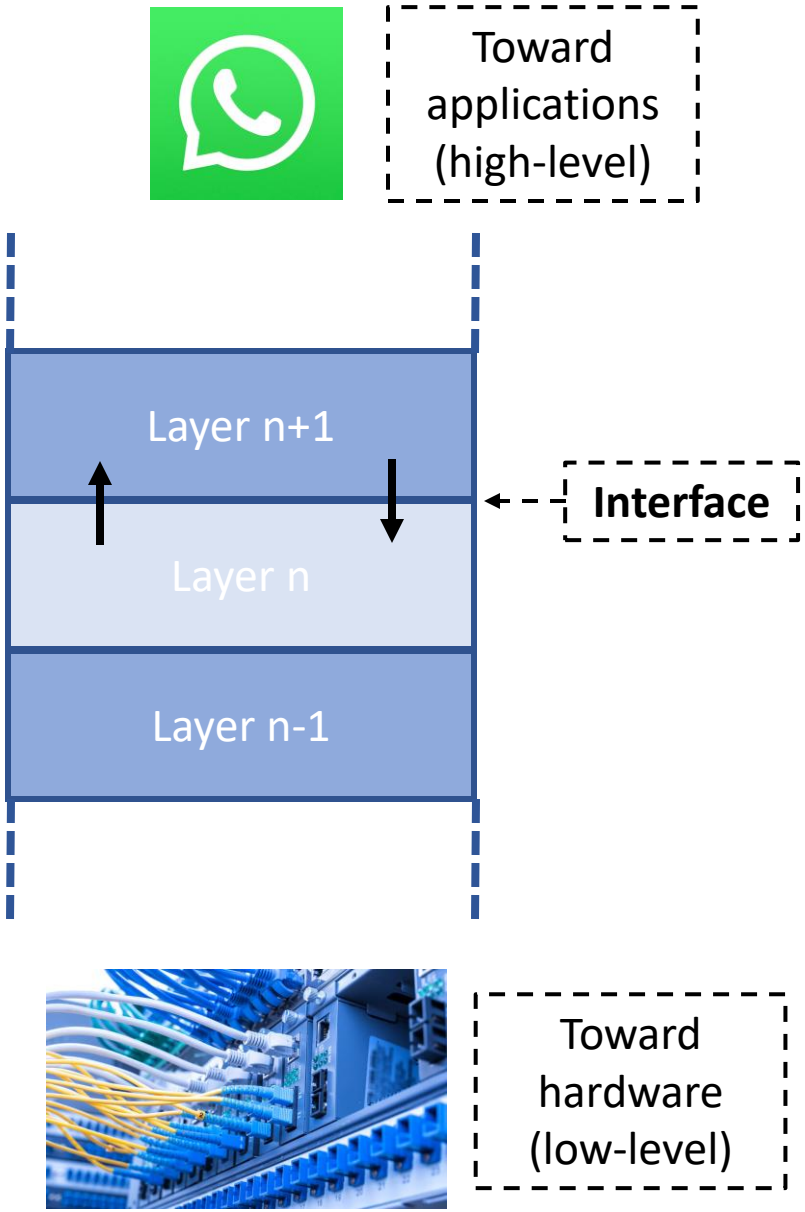
- Each one of these working groups produces numbered documents called **Request For Comments** (RFC) including description and definition of protocols, concepts, methods underlying a standard.
  - For example, the IPv4 protocol was defined in RFC 791 (1981).

Introduction

Network Models: Layered Model

- A reasonable approach to face the different communication problems is to design a **layered model** (*Divide et Impera*):
  - Each layer is conceptually **responsible for one specific task** (solves one problem).
  - Each layer rely on services from the **lower layer** and provides services to the **upper layer**.

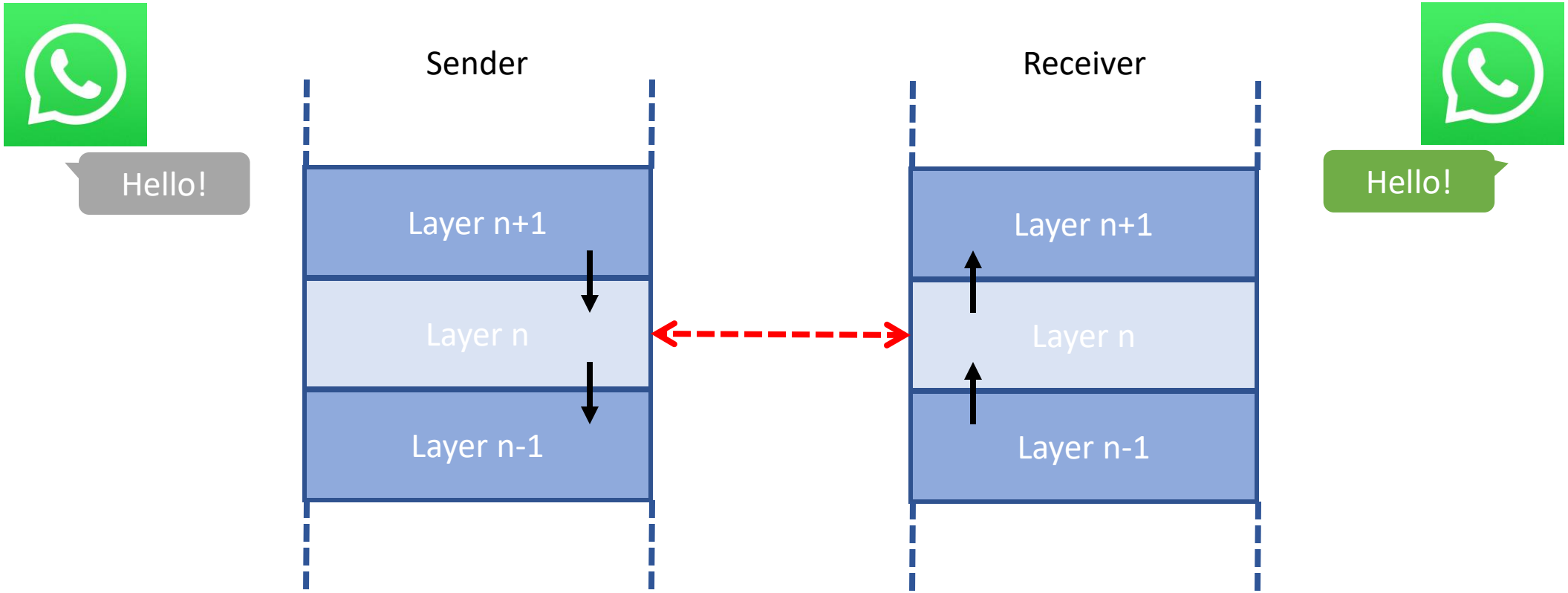
- Pros:
  - **Modularity**: layers are simple and independent
- Cons:
  - **Scalability**: climbing too many layers is inefficient



Introduction

Network Models: Layered Model

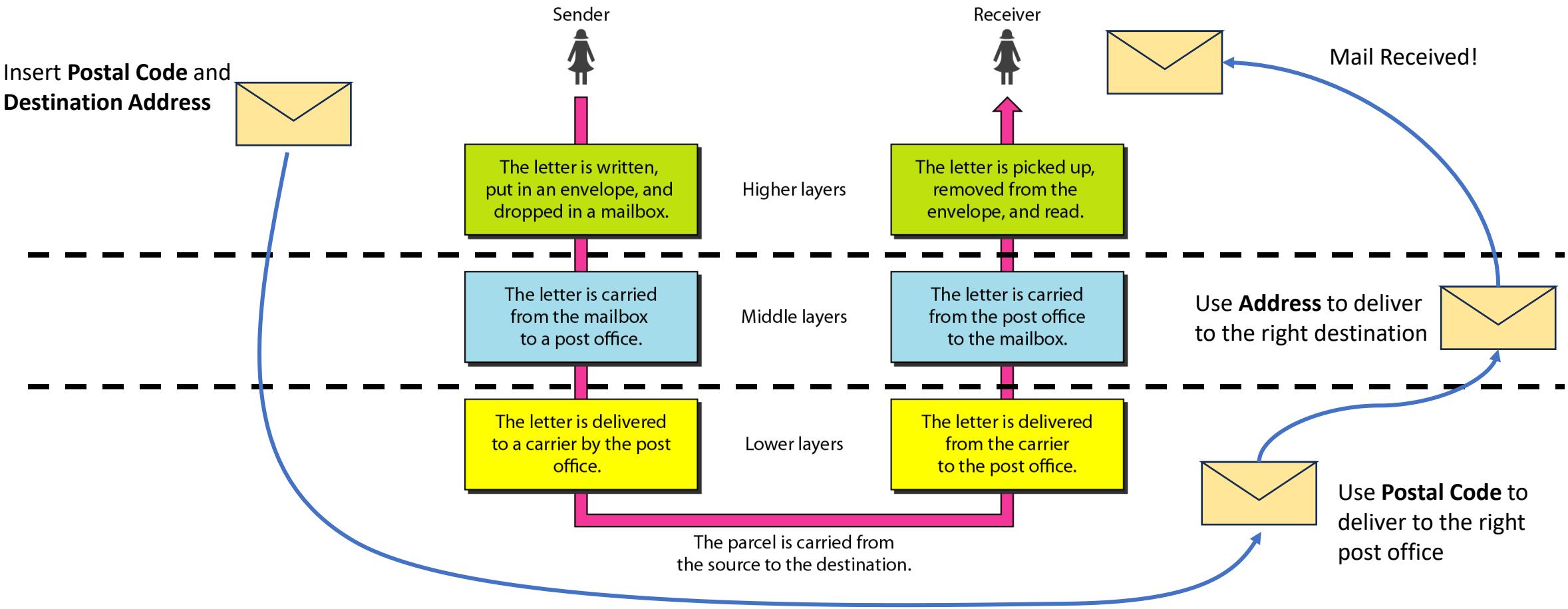
- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.



Introduction

Network Models: Layered Model (example)

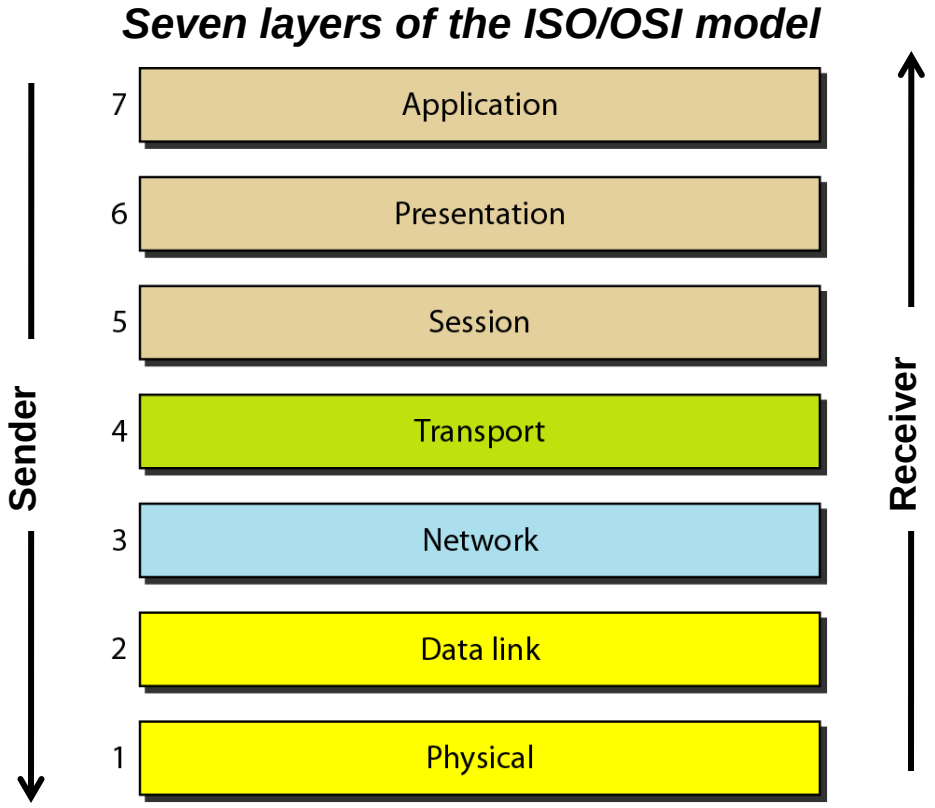
- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.



Introduction

Network Models: The OSI Model

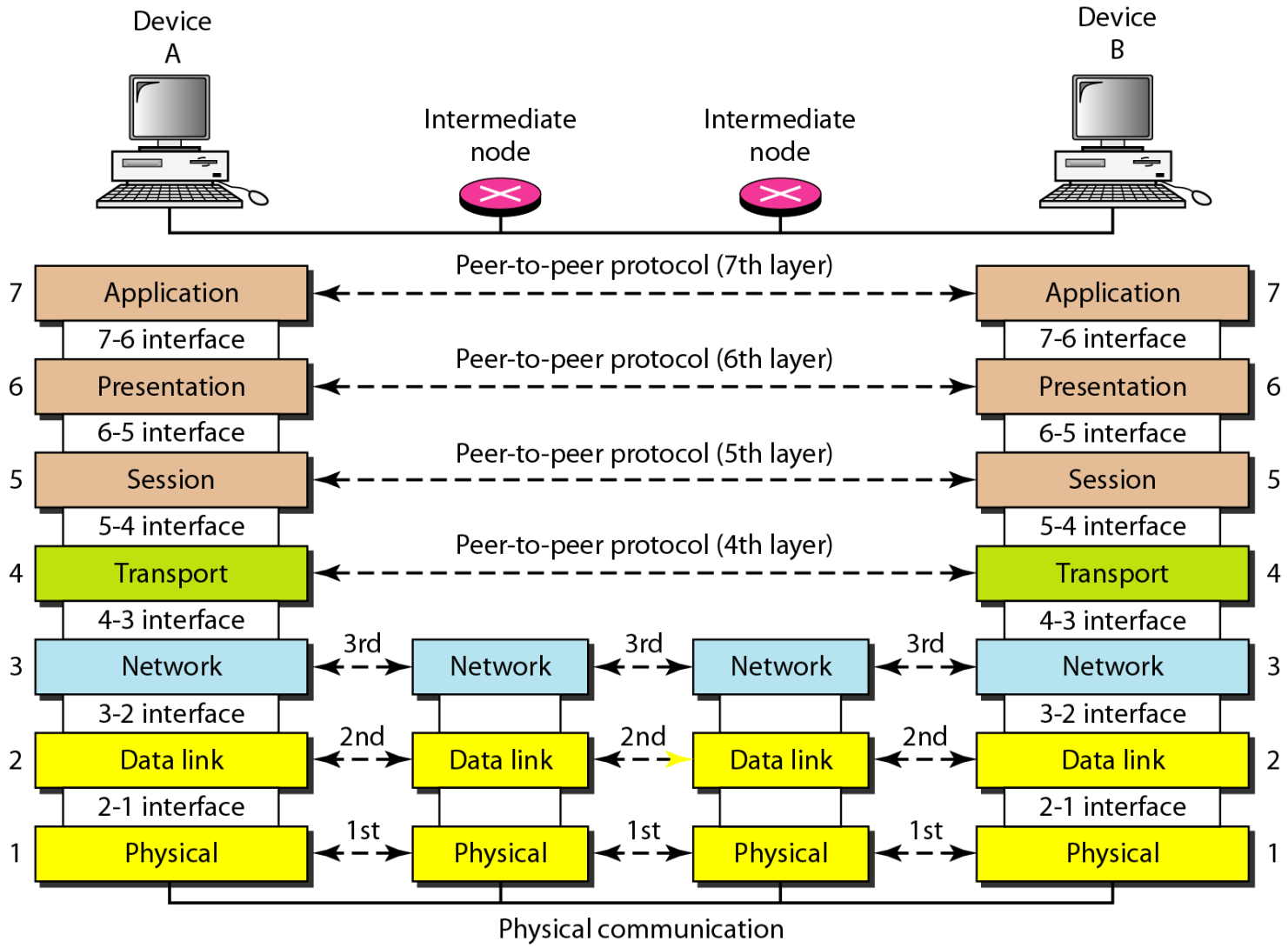
- In the ‘80 the **ISO** (International Standards Organization) defined a layered model for computer networks: the **OSI** (Open System Interconnection) model.
- The ISO/OSI model includes 7 layers:
  1. Physical transmission of raw bits.
  2. Data format definition (frames).
  3. Routing of messages through the net.
  4. Transmission protocols (TCP, UDP).
  5. Management of ports and sessions (continuity of data stream).
  6. Translation of messages (encoding, compression, decryption, etc.).
  7. Data use and human-computer interaction (file sharing, emails, streaming videos, etc.).



Introduction

Network Models: The OSI Model

- Not always all layers are implemented.
- Typically, in intermediate nodes (switches, routers), **only 2 or 3 layers** are implemented (media layers).
- Only the **end-points implement the whole stack** up to the application layers.

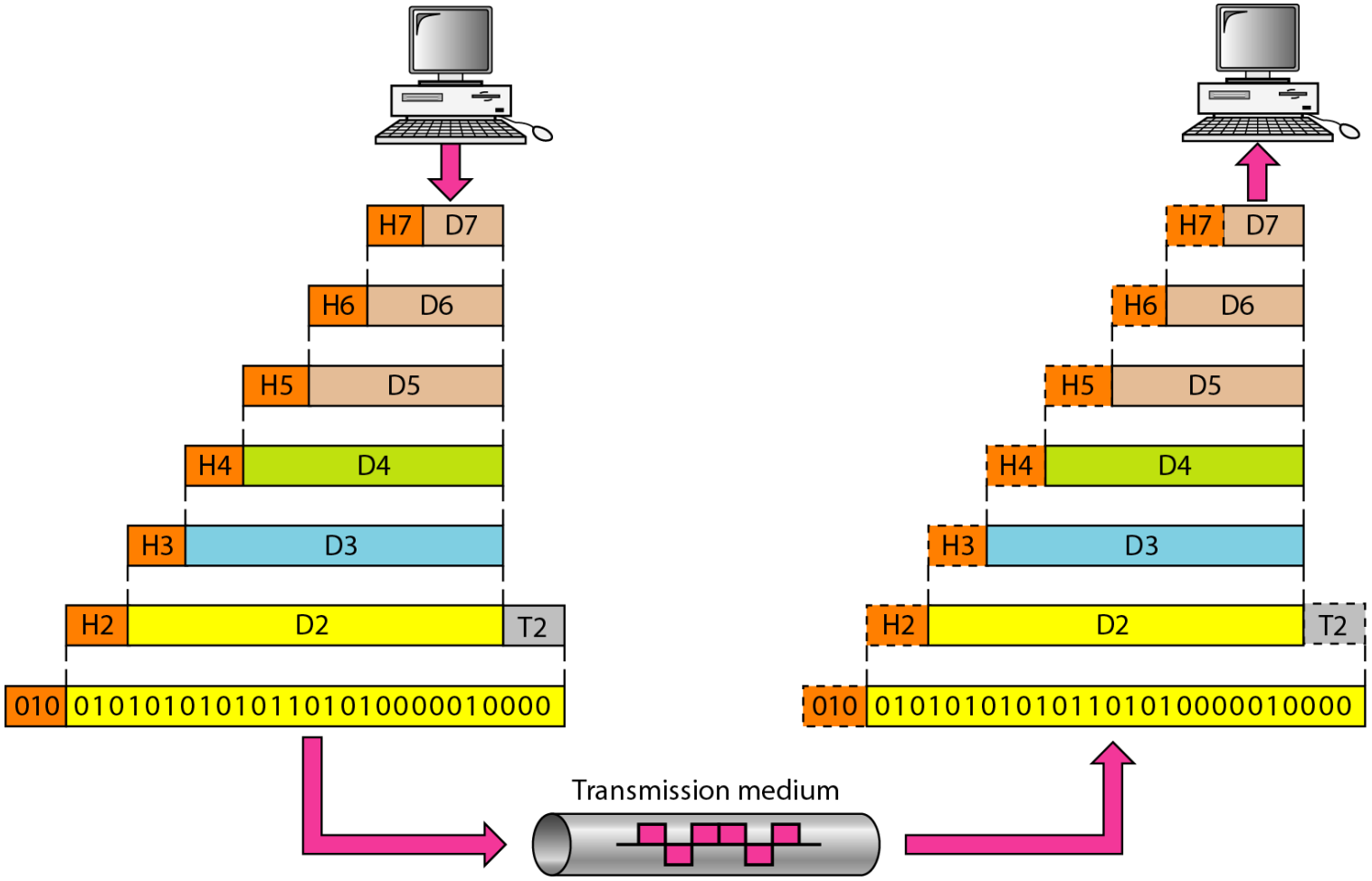


Introduction

Network Models: The OSI Model

- **Encapsulation** (top-down): every layer of the sender adds a layer-specific field to the message (payload) in the form of a header (H) or trailer (T).

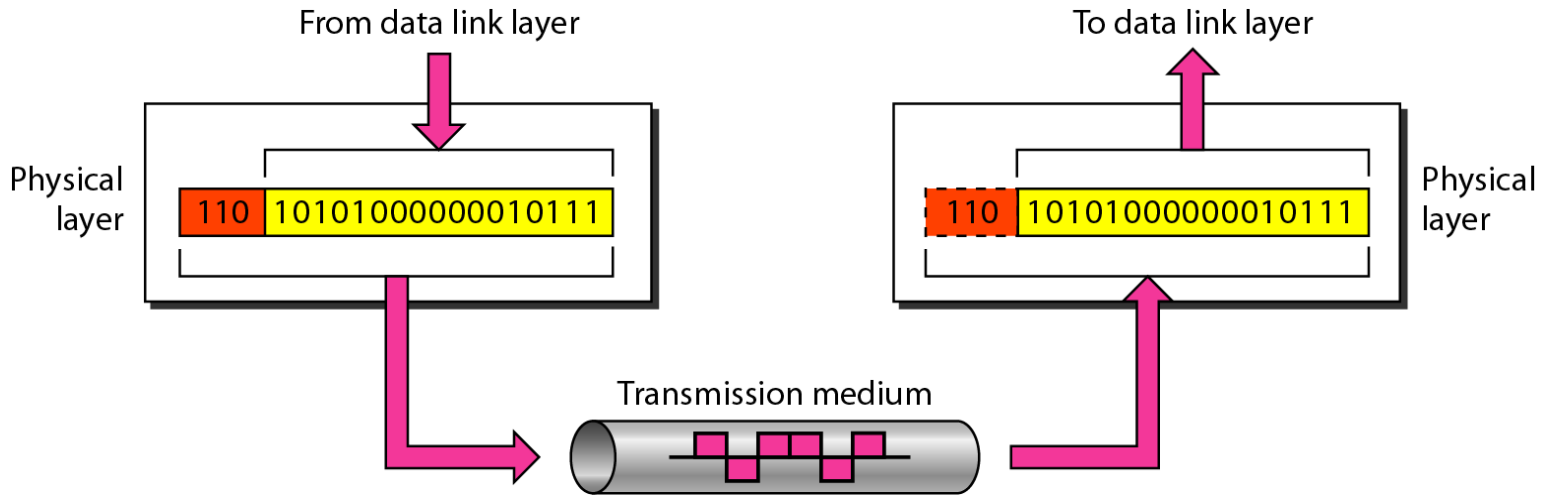
- **Decapsulation** (bottom-up): Those fields are removed and interpreted by the same layer in the receiver.



Introduction

OSI Model: 1. Physical Layer

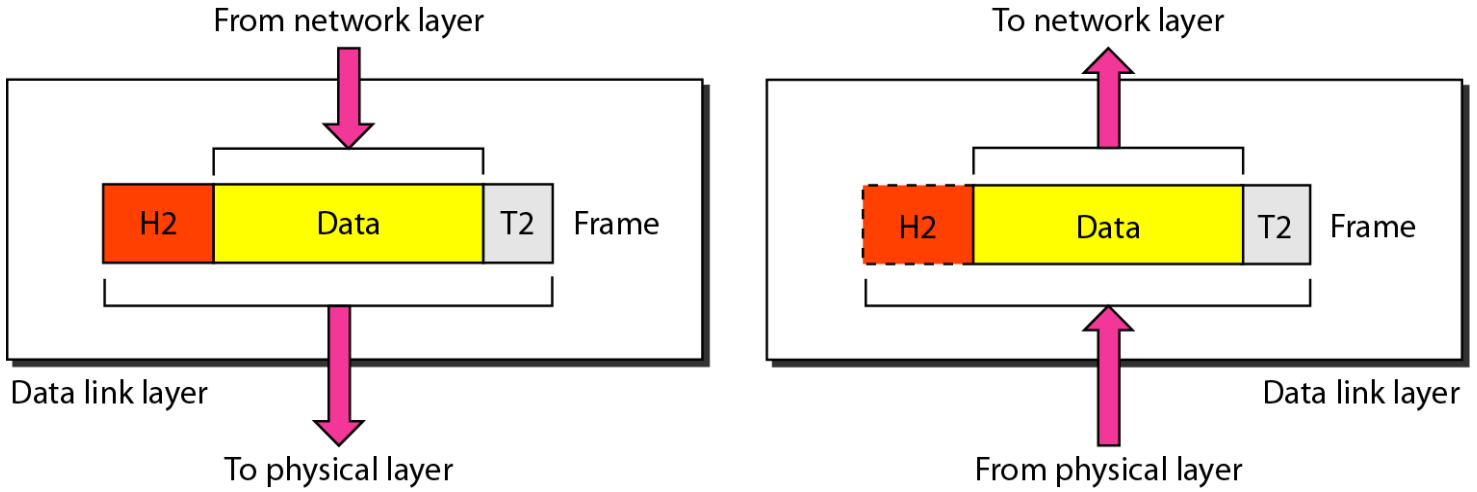
- The **physical layer** is responsible for movements of individual bits from one hop (node) to the next.
  - Physical characteristics of interfaces and media (connectors, cables, electric signals)
  - Line configuration (point-to-point or multipoint)
  - Physical topology (mesh, star, ring or bus)
  - Transmission mode (simplex, half-duplex or duplex)
  - Representation of bits
  - Data rate
  - Synchronization of bits



Introduction

OSI Model: 2. Data Link Layer

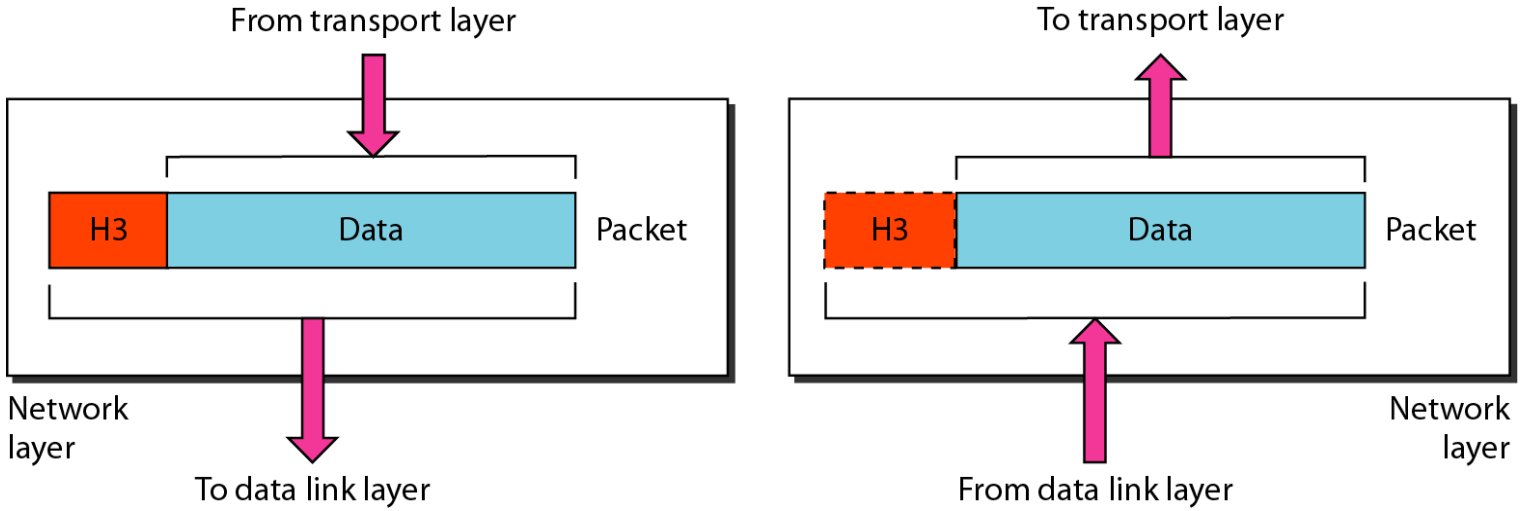
- The **data link layer** is responsible for moving frames from one hop (node) to the next (segment).
  - Works on frames (portion of data, typically few hundreds of bytes)
  - Flow and error control (frame control sequences)
  - Access control



Introduction

OSI Model: 3. Network Layer

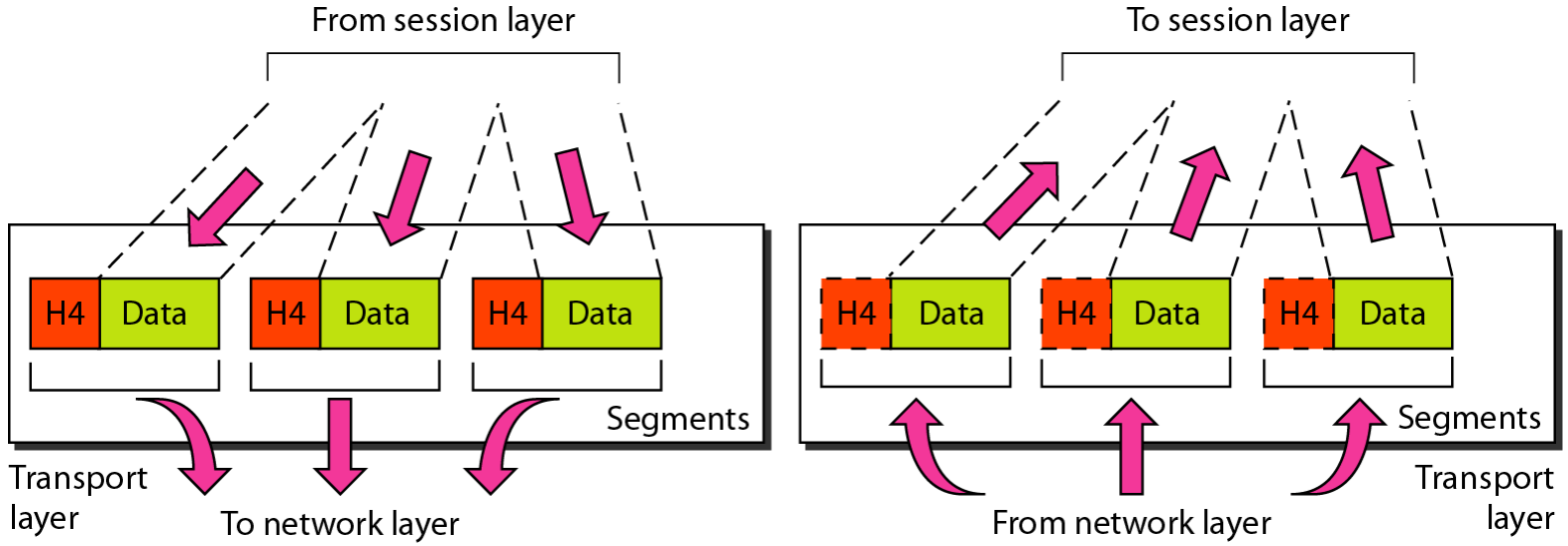
- The **network layer** is responsible for the delivery of individual packets from the source host to the destination host (path).
  - Works on packets (typically larger and more complex than frames)
  - Source-to-destination delivery: packets from the source to the destination.
  - Logical addressing
  - Routing (routing tables)



Introduction

OSI Model: 4. Transport Layer

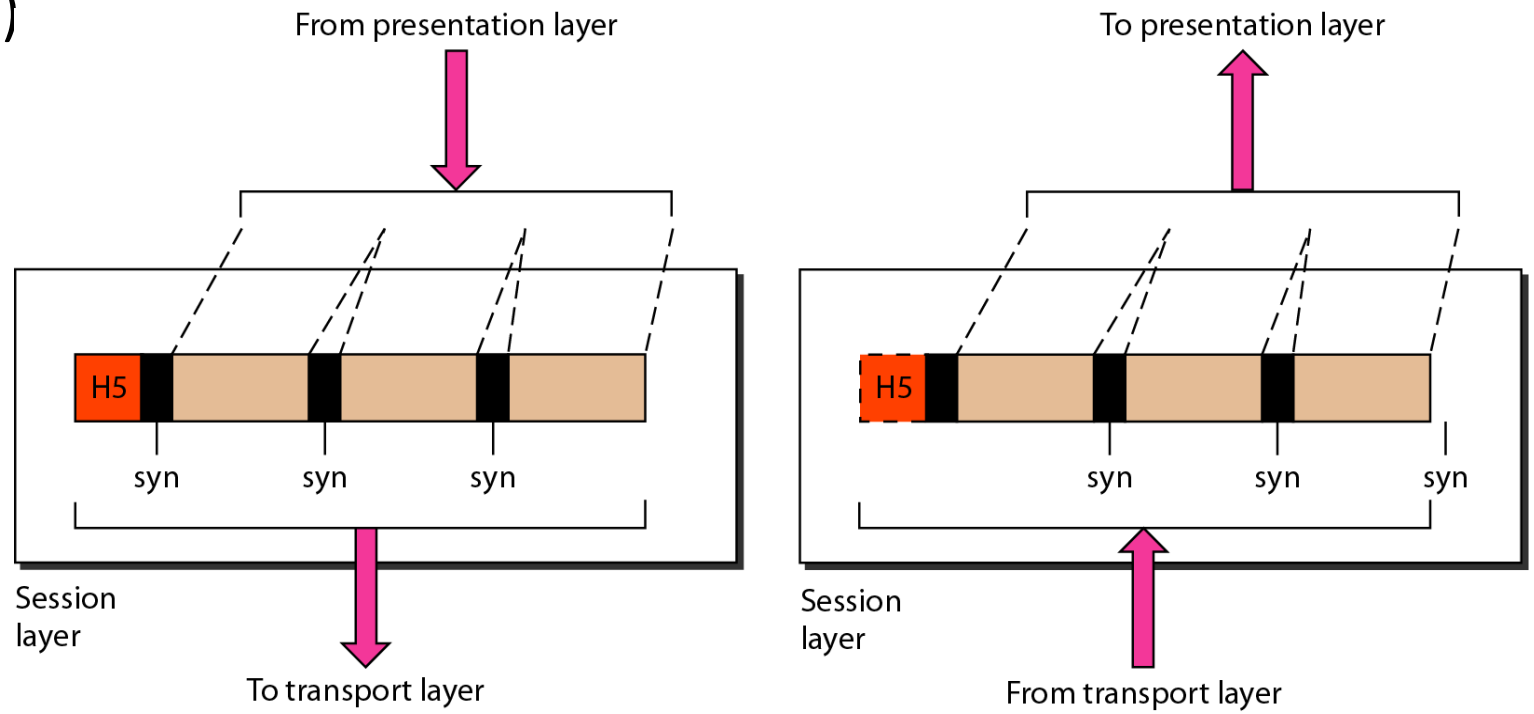
- The **transport layer** is responsible for the delivery of a message from one end to another (message is received by the right program).
  - End-to-end delivery
  - Connection control (Connection-oriented or connection-less)
  - Segmentation/reassembly (to/from packets of layer 3)
  - Port addressing
  - Flow and error control



Introduction

OSI Model: 5. Session Layer

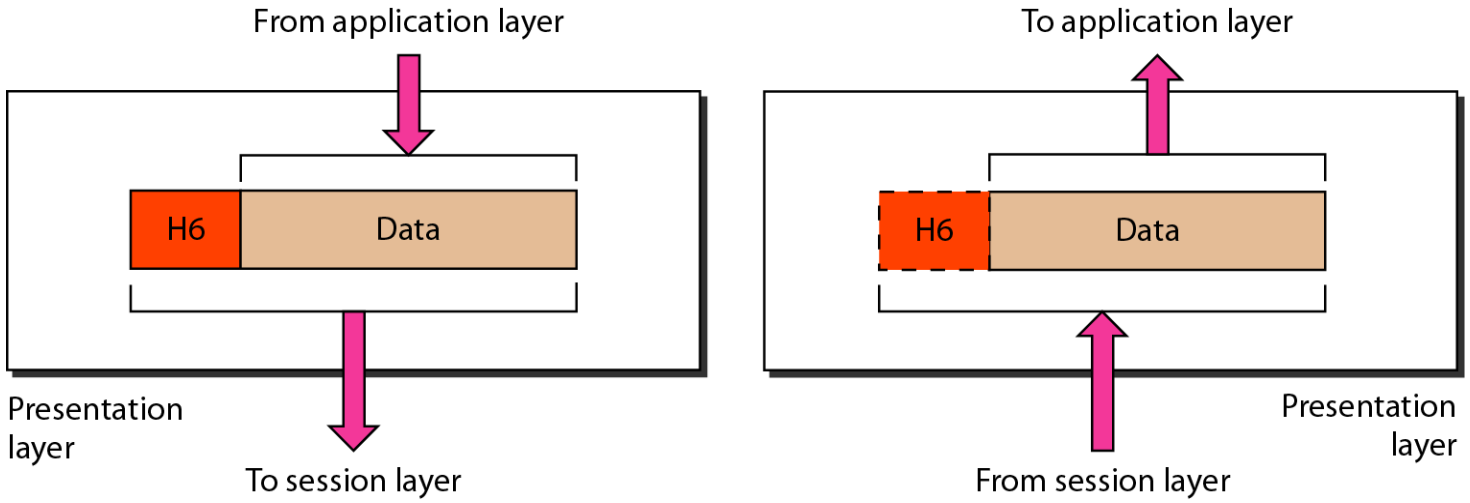
- The **session layer** is responsible for dialog control and synchronization of request/response.
  - It establishes, maintains and synchronize the interaction between communicating system (communication session).
  - Dialog control and management
  - Synchronization (checkpoints)



Introduction

OSI Model: 6. Presentation Layer

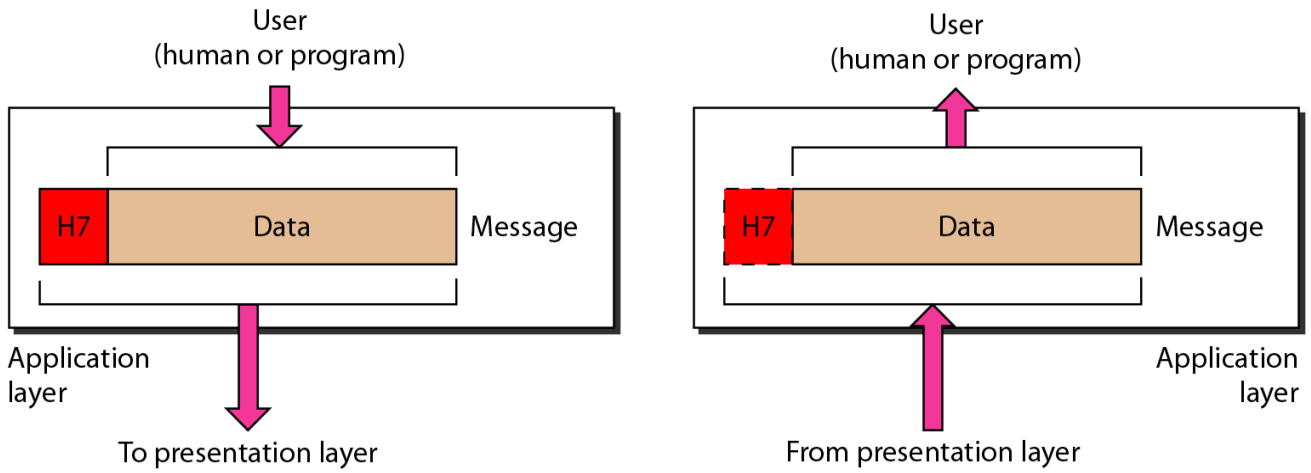
- The **presentation layer** is responsible for the representation of data, i.e., translation, compression, encryption, etc.
  - Translation (e.g., EBCDIC-coded or ASCII-coded to text)
  - Encryption and Decryption
  - Compression



Introduction

OSI Model: 7. Application Layer

- The **application layer** is responsible for providing services to the user.
  - Network virtual terminal (Remote log-in)
  - File transfer and access
  - Mail services
  - Accessing the World Wide Web
  - …



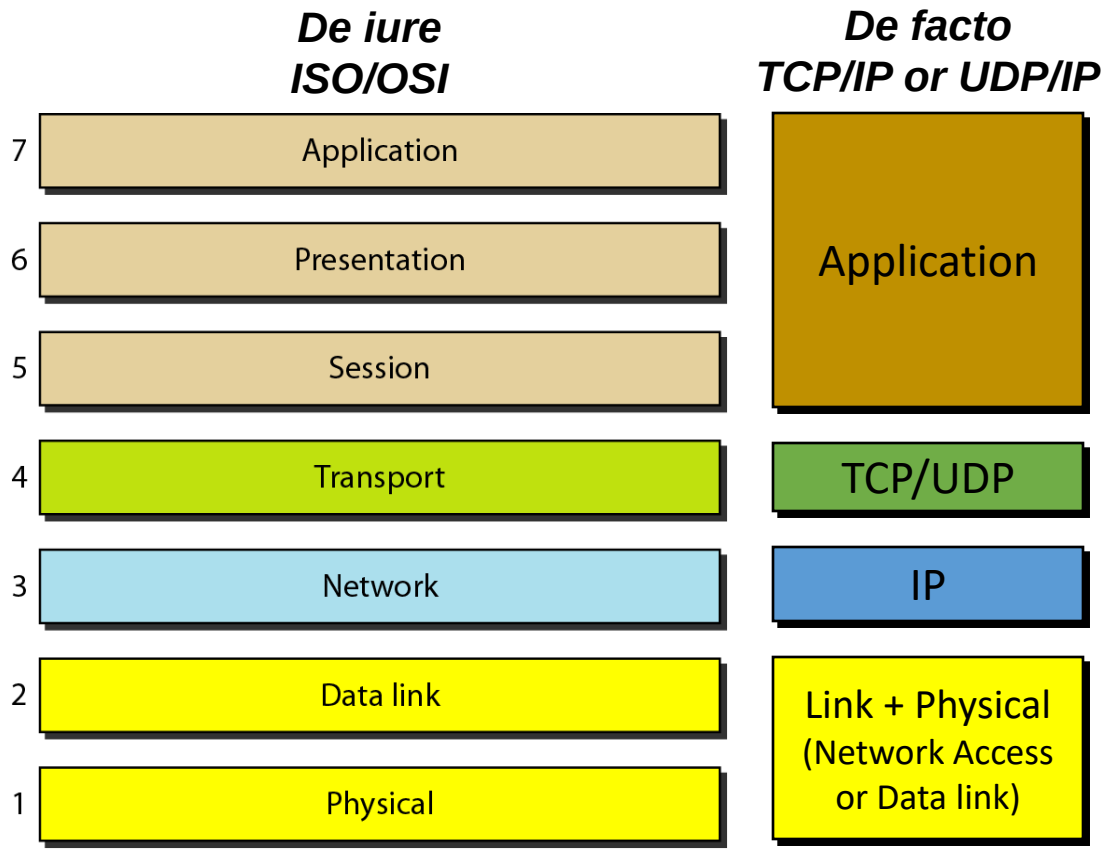
Introduction

Network Models: The TCP/IP Model

- The ISO/OSI model is *de iure* the standard stack for computer networks, but it is quite complex and detailed.

- The TCP/IP and UDP/IP (aka Internet Protocol Suite) are the set of communication protocols **actually used** on internet and local networks.
  - TCP: Transmission Control Protocol.
  - UDP: User Datagram Protocol.
  - IP: Internet Protocol.

- The TCP/IP and UDP/IP are *de facto* the standard stack for internet communication.



Introduction

Network Models: Other Protocols

- There are different protocols for device communication, which can be used depending on the situation, that are still relying on the layered model.

